

Software Testing

Testing is the process of executing a program with the intention of finding errors.

Testing improve the software in terms of efficiency, accuracy, and usability.

Testing = Verification+ Validation

Verification: it refers to the set of tasks that ensure that the software correctly implements a specific function.

Validation: it refers to a different set of tasks that ensure that the software that has been built is traceable to customer requirements.

Verification: “Are we building the product right?”

Validation: “Are we building the right product?”

Difference between Testing and Debugging

Testing	Debugging
Testing is the process of executing a program with the intention of finding errors.	<u>Debugging</u> is the process of correcting the bugs found during testing.
It is the process to identify the failure of implemented code.	It gives solution to code failure.

Testing	Debugging
Testing is the display of errors.	Debugging is a deductive process.
Testing is done by the tester.	Debugging is done by developer.
There is no need of design knowledge.	Debugging can't be done without proper design knowledge.
Testing can be done by insiders as well as outsiders.	Debugging is done only by insiders.
Testing can be manual or automated.	Debugging is always manual.
It is based on different testing levels i.e. unit testing, integration testing, system testing, etc.	Debugging is based on different types of bugs.
Testing is a stage of the software development life cycle (SDLC).	Debugging is not an aspect of SDLC, it occurs as a consequence of testing.
Testing is initiated after the code is written.	Debugging commences with the execution of a test case.
Testing process based on various levels of testing.	Debugging process based on various types of bugs.

Who tests the software

Quite a few different people and roles participate in testing process. They include:

- **Software Testers.**
- **QA Analysts.**
- **Acceptance Testers.**
- **Test Managers.**
- **Test Engineers.**
- **QA Testers.**
- **Test Analysts.**
- **SQA.**

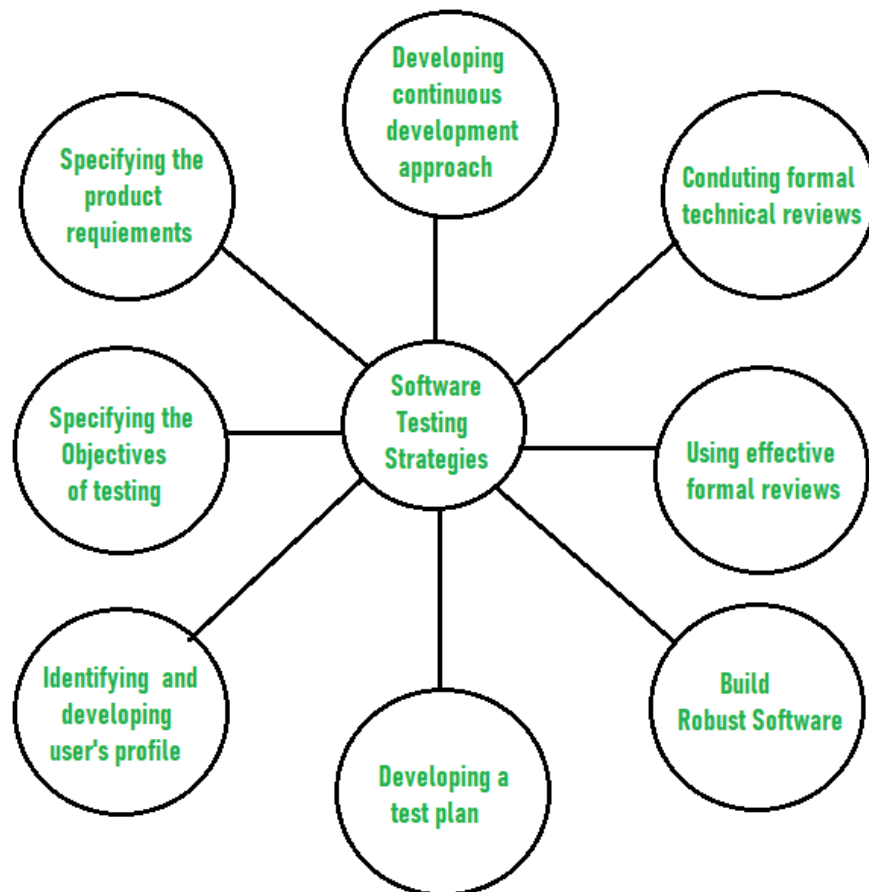
Software Testing Strategies

The following are common testing strategies:

1. Black box testing – Tests the functionality of the software without looking at the internal code structure.
2. White box testing – Tests the internal code structure and logic of the software.
3. Unit testing – Tests individual units or components of the software to ensure they are functioning as intended.
4. Integration testing – Tests the integration of different components of the software to ensure they work together as a system.
5. Functional testing – Tests the functional requirements of the software
6. System testing – Tests the complete software system to ensure it meets the specified requirements.

7. Acceptance testing – Tests the software to ensure it meets the customer's or end-user's expectations.
8. **Regression testing** – Tests the software after changes or modifications have been made to ensure the changes have not introduced new defects.
9. Performance testing – Tests the software to determine its performance characteristics such as speed, scalability, and stability.
10. Security testing – Tests the software to identify vulnerabilities and ensure it meets security requirements.

The overall strategy for testing software includes:



Advantages of software testing:

- Improves software quality and reliability
- Enhances user experience
- Increases confidence – By testing the software, developers and stakeholders can have confidence that the software meets the requirements and works as intended.
- Facilitates maintenance.
- Reduces costs

Disadvantages of software testing:

- Time-consuming
- Resource-intensive
- Unpredictable results
- Delays in delivery.

Types of software testing

1. Manual Testing:

Manual testing includes testing software manually, i.e., without using any automation tool or any script.

Testers use test cases to test software.

Examples of Manual tools are **Jira, LoadRunner, Citrus, etc.**

2. Automation Testing:

In automation testing, automation tools are used to execute the test cases. It is much faster than the manual testing.

Hence, we can enhance the speed of our test execution.

Examples of Automation testing tools are **Selenium, Cucumber, Cypress etc.**

White box testing (Structural Testing)

[White Box Testing](#) is a software testing method in which the internal structure/design of the item being tested is known to the tester. Implementation and impact of the code are tested.

This is called White box/ clear box as the code is visible to developer.

Black box testing (Functional Testing)

[Black Box Testing](#) is a software testing method in which the internal structure/design of the item being tested is not known to the tester. Only the external design and structure are tested.

Gray box testing

Gray box testing is a combination of white box and Black box testing. It can be performed by a person who knew both coding and testing.

Black Box Testing	White Box Testing
Here the functionality of software is tested without knowing its internal structure or coding.	In white-box testing, the internal structure of the software is known to the tester.
Black Box Testing is also known as functional testing, data-driven	It is also known as structural testing, clear box testing, code-based testing,

Black Box Testing	White Box Testing
testing, and closed-box testing.	and transparent testing.
It is mostly done by software testers.	It is mostly done by software developers.
Less programming knowledge is required.	Programming knowledge is required.
It can be referred to as outer or external software testing.	It is the inner or the internal software testing.
It is a functional test of the software.	It is a structural test of the software.
It is done at higher levels of testing (System testing and acceptance testing).	It is done at lower levels of testing (Unit testing and integration testing)
It is less time-consuming.	It is more time-consuming.
It is the behavior testing of the software.	It is the logic testing of the software.
Black Box Testing types: • Regression Testing • Functional Testing	White Box Testing types: • Condition Testing • Path Testing

Black Box Testing

- Non-Functional Testing

It is not suitable for algorithm testing.

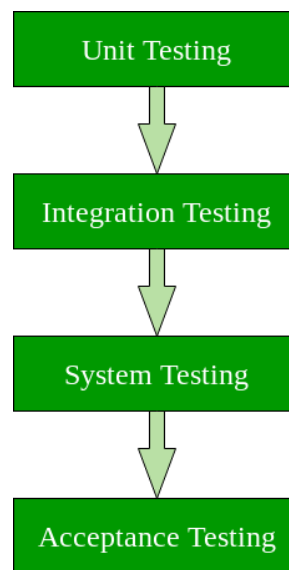
White Box Testing

- Loop Testing

It is suitable for algorithm testing.

Levels of Testing

Software testing can be classified into **4 levels**:



1. Unit Testing:

A level of the software testing process where individual units/components of a software are tested. The purpose is to validate that each unit of the software performs as designed.

2. Integration Testing:

A level of the software testing process where individual units are combined and tested as a group. The purpose of this level of testing is to expose faults in the interaction between integrated units.

3. System Testing:

A level of the software testing process where a complete, integrated system/software is tested. The purpose of this test is to evaluate the system's compliance with the specified requirements.

4. Acceptance Testing:

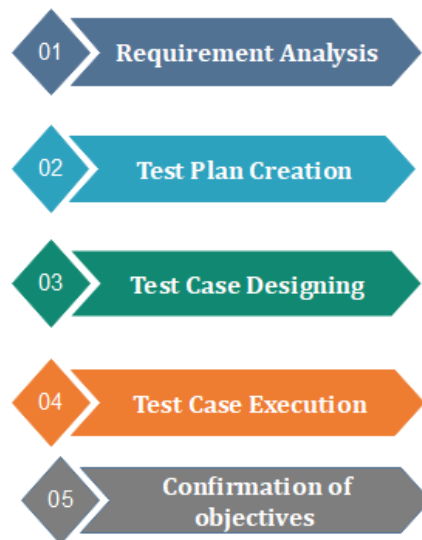
It is a black-box testing process where the functionality is verified to ensure the software meets the acceptance criteria.

It is the last phase of software testing.

Types of Acceptance testing:

- User Acceptance Testing
- Alpha Testing
- Beta Testing

User acceptance testing (UAT) is a type of testing, which is done by the customer before accepting the final product.



Alpha Testing	Beta Testing
Alpha testing is performed by Software Testers.	Beta testing is performed by end users
Alpha testing involves both the white box and black box testing.	Beta testing mainly involves black-box testing.
Alpha testing is performed at the developer's site.	Beta testing is performed at users location.
Reliability and security testing are not checked in alpha testing.	Reliability, security and robustness are checked during beta testing.
Alpha testing requires a testing environment or a lab.	Beta testing doesn't require a testing environment or lab.
Alpha testing may require a long execution cycle.	Beta testing requires only a few weeks of execution.
Developers can immediately address the critical issues or fixes in alpha testing.	Most of the issues are implemented in future versions of the product.
Multiple test cycles are organized in alpha testing.	Only one or two test cycles are there in beta testing.

Q: Design a test case for testing a login form module.